Pooja Ravi

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EDUCATION

MSc, Applied Computing 2023 - 2025
University of Toronto, Canada
Awards: Vector Scholarship in Artificial Intelligence (2023-24) | Mitacs Globalink Graduate Fellowship

BTech, Computer Science and Engineering 2019 - 2023
SRM Institute of Science and Technology, India

RESEARCH EXPERIENCE

Researcher | University of Toronto (TCIG) Mar 2026 - Present
Supervised by David B. Lindell

Gripper-object Manipulation

- **Contact sensing:** Vision-based estimation of gripper-object contact locations and 3-axis force loads from tracked deformation, without tactile or controller inputs at inference.
- **Compliant gripper design:** Exploring visible bending and recovery in UMI-style grippers so camera-observed deformation can support contact-guided manipulation across grippers and objects.

In-hand Robotic Object Manipulation

- **Action recovery:** Tracked ray-traced hand points with CoTracker; recovered joint angles by reprojection-error inverse kinematics, warm-started from prior frames, then refined with constrained cross-entropy method (CEM) search for intended motion.
- **Physics replay:** Executed recovered actions in MuJoCo with contact and gravity; passed a frame from each replay back as the next video-generation cycle's initial frame, closing the loop.
- **Closed-loop pipeline:** Linked a fine-tuned object-motion-conditioned video model to CoTracker, inverse kinematics, constrained CEM, and MuJoCo feedback for ORCA Hand v1, yielding executable joint-space actions.

ProxyPose (NeurIPS 2026)

- **Benchmarking:** Engineered a unified 6-DoF benchmark across HO3D, YCBinEOAT, and YCB-Video; standardized meshes, RGB/depth frames, pose annotations, and metadata in a consistent evaluation format.
- **Evaluation:** Implemented metrics tracking temporal rotational and translational drift; built evaluation pipelines, ran competing methods, and analyzed cross-dataset results.

PUBLICATIONS

1. **ProxyPose: 6-DoF Pose Tracking via Video-to-Video Translation**
   Ruihang Zhang, Felix Taubner*, **Pooja Ravi**, Kiriakos N. Kutulakos, and David B. Lindell*
   Advances in Neural Information Processing Systems (**NeurIPS 2026**). Paper. * Equal contribution.

2. **Attention Mechanism, Linked Networks, and Pyramid Pooling Enabled 3D Biomedical Image Segmentation**
   ***Pooja Ravi**, Srijarko Roy, Indira Dutta, and Kottilingam Kottursamy*
   2022 IEEE/ACIS 23rd International Conference on Software Engineering, Artificial Intelligence, Networking and Parallel/Distributed Computing (**SNPD 2022**), pp. 91-96. Paper.

3. **Real-time Multi-module Student Engagement Detection System**
   ***Pooja Ravi** and M. Ali Akber Dewan*
   In Communication and Intelligent Systems: Proceedings of **ICCIS 2022**, Lecture Notes in Networks and Systems, vol. 686, pp. 261-278, Springer, 2023. Paper.

INDUSTRY EXPERIENCE

Machine Learning Engineer | Q2 Software May 2024 - Mar 2026

- Led cheque-signature verification across 200K images, training variational autoencoder embeddings and Siamese networks with triplet and contrastive losses; fine-tuned multimodal LLMs for pairwise visual comparison.
- Trained LayoutLM, Donut (Swin+BART), and TrOCR on 100K+ cheques; improved text extraction by 8% and fraud flagging by 4% while reducing inference latency through model optimization.

Machine Learning Engineer | 16 Bit Oct 2023 - Mar 2024

- Led X-ray translation across 12-96 month follow-ups to model knee osteoarthritis progression; scaled GAN/diffusion training on 70K+ images with PyTorch Lightning and LoRA, achieving ~10x faster training and ~19% higher image realism.

SELECTED PROJECTS

Echo Charlie: multimodal voice restoration pipeline; Boson AI Hackathon winner (2025).

2D Sketch to 3D Stylization: Converts 2D sketches into stylized 3D views using image style transfer and NeRFs.